

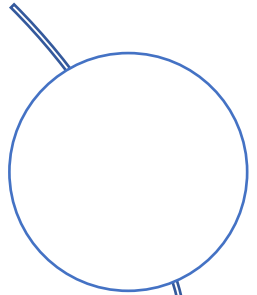
BLURRING THE INTERNAL-EXTERNAL DISTICTION

ANNOUNCEMENT

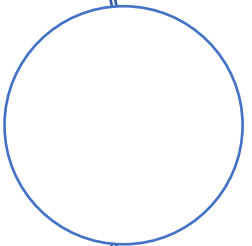
Assignment: Role Play of a Scientific Figure

Deliverable 3: Full Submission – due Feb 15th, 11:59 pm

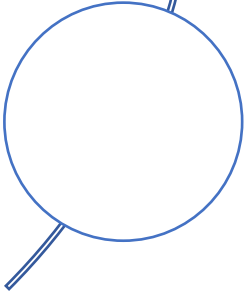
The Plan!



Recap



**Blurring the Internal-External
Distinction**



In-Class Activity

RECAP

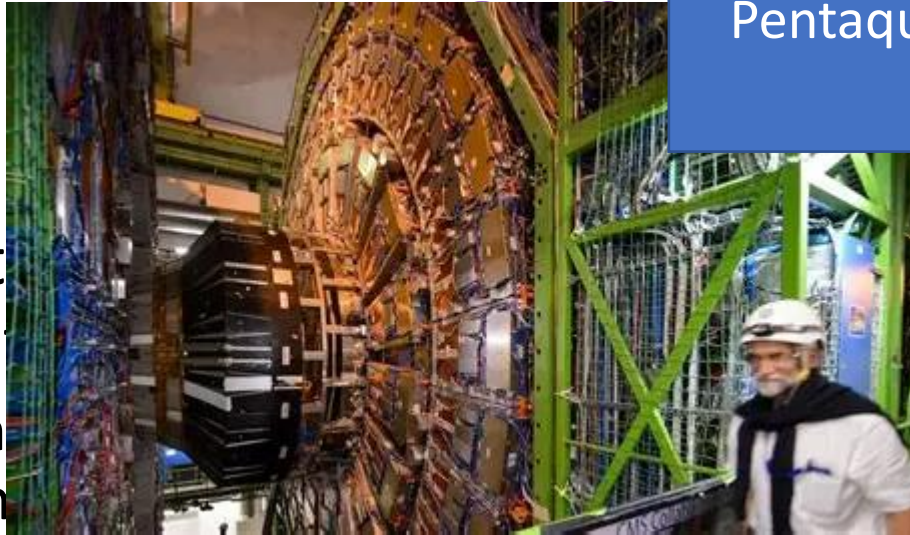


OBSERVATION

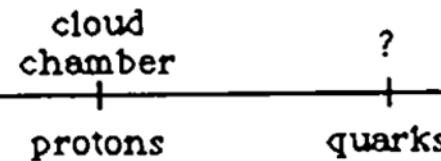
- Observation is the touchstone of ***external virtue*** – it is the contact of a theory to the outside world.
 - The underlying expectation – when we discussed explanation and confirmation – was that observation is *a source of objectivity in science*.
- ***However, this needs a closer investigation***
 - ... we just saw earlier that there are degrees of observability parallel to degrees of indirectness in the observation!

Last time!

- Observat
a theory
- The un
confirm



Pentaquarks have been observed at CERN.



contact of

ience.

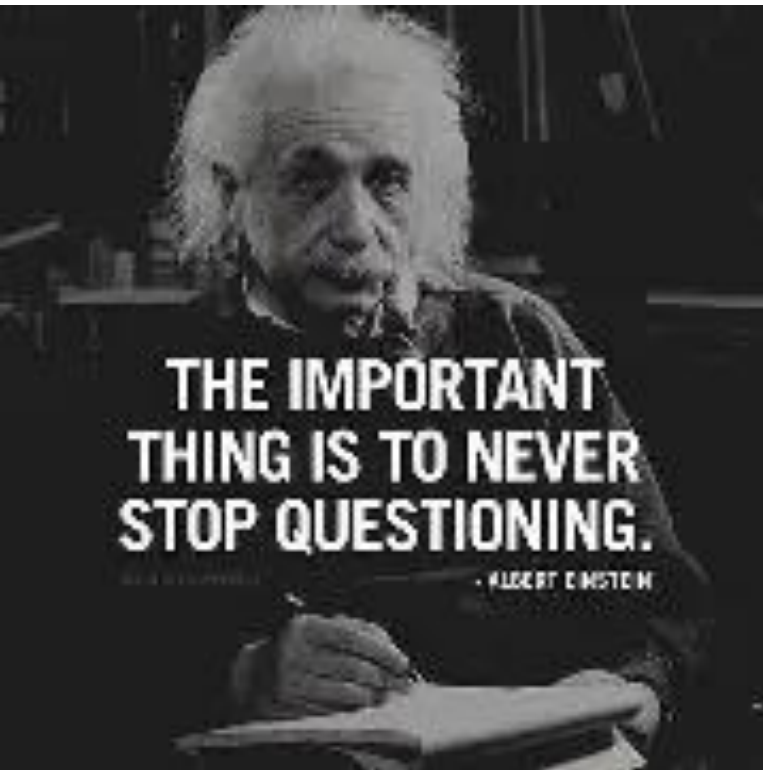
- ***However, this needs a closer investigation***

- ... we just saw earlier that there are degrees of ob
degrees of indirectness in the observation!

But the search for the top
quark continues ...
The search for 'truth'
continues



OBSERVATION



- But are these examples of information (e.g., cloud chamber images, the Hubble telescope, etc.) straight from nature, free of any conceptual embellishment, the undeniable, impartial, common ground of objectivity?

*Asking this question is important!
Why?*



OBSERVATION

- Because we deal with *empirical sciences*: natural sciences (e.g., physics, biology) and social sciences (e.g., psychology and anthropology)
 - They all make claims on what is going on in the actual world (the world of elements, organisms, environments, etc.)
 - They contrast with claims in nonempirical sciences (e.g., mathematics, logic), which can be assessed without having to look at what's happening in the world.
 - For instance, I don't need to look outside in order to know that $2 + 2 = 4$ or that a triangle has three sides.



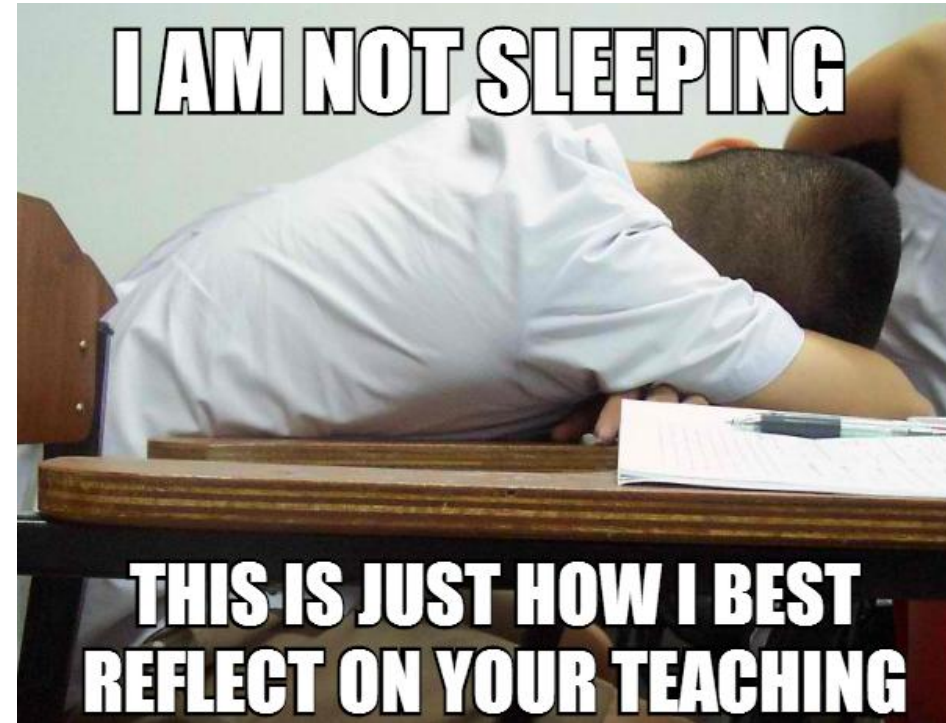
OBSERVATION

- The (empiricists') hope is that we all agree on observations since observations are accessible to all of us.
 - Even if we are skeptical about a theory or lean towards an alternative theory, we can all agree on the observations.
 - ***Uhhhh Ohhhh*** so observations are free of inference and interpretations and, thus, there is no danger of error and/or bias?

OBSERVATION



- Consider the following claims:
 - There are 15 people in the room.
 - One person is asleep.
 - 32 people have taken notes of the assigned readings.



These claims are easy to check, and we'll reach an agreement easily...

OBSERVATION



- But how about those claims:
 - There is at least one cosmic ray in the room at any one moment.
 - On average there are more cosmic rays going down than there are going horizontally.
 - There are fewer cosmic rays in the room on Sunday than on any other day.



Which of these questions can be checked and agreed on?



OBSERVATION

- But how about those claims:

However, the agreement disappears as long as there is an alternative theory.

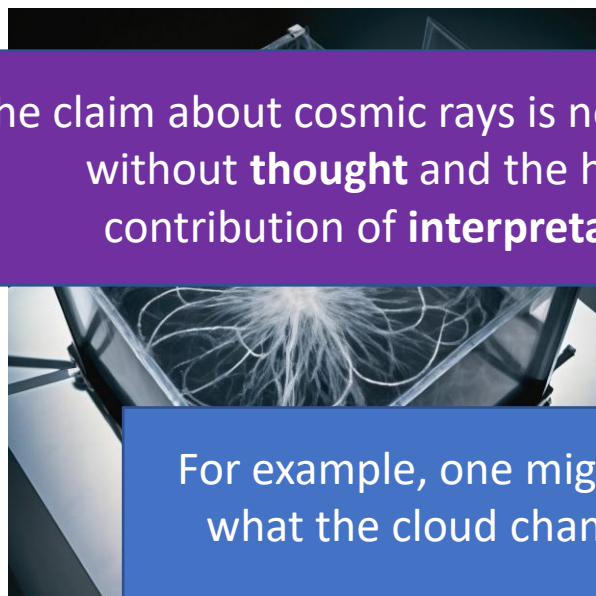
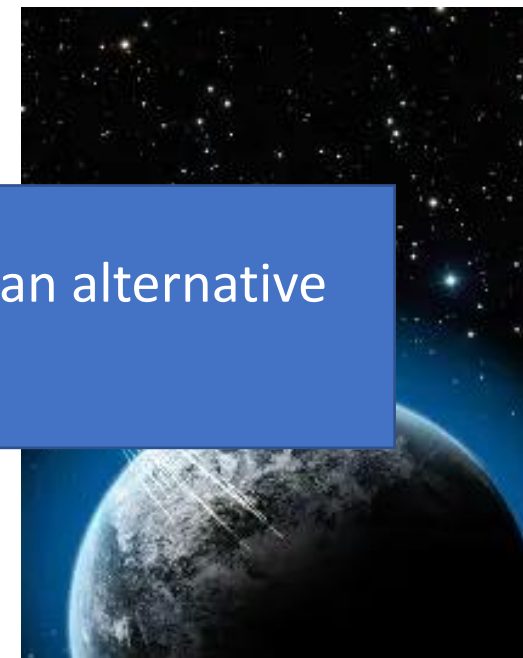
The claim about cosmic rays is not decidable without **thought** and the human contribution of **interpretation**.

cosmic rays
going

cosmic rays in
than on any other day

Which of these questions can be checked and agreed on?

For example, one might suggest that the lines are formed by a magnetic effect and that what the cloud chamber shows are actually the magnetic field lines of the earth and nearby magnets ...



OBSERVATION



- Thus, the focus of our understanding of observation must be on observational report rather than on the physical event of sensation.

OBSERVATION

- **Accountable Observation:**

- To be useful as an evidence, an observation must make contact with a theory in some way if it is to serve as proof or as refutation.
- Science can make use of observation only if it is an accountable observation:
 1. The observation must be *informative* in the sense that it must be an account of something if it is to have the assertive content required to serve as justification of other claims.
 - It requires a claim as to **what** is observed (*What do you see?*) and an assertion of the **information** that is being conveyed to the world (*Do you see that this is a that?*)
 - If an observation report is to be relevant to a theory it must include some of the language of the theory.
 - It must be assertively be evidence of an x if it is to be evidence in the case to justify a theory about x.



OBSERVATION

- **Accountable Observation:**

- To be useful as an evidence, an observation must make contact with a theory in some way if it is to serve as proof or as refutation.
- Science can make use of observation only if it is an accountable observation:
 1. The observation must be *informative* in the sense that it must be an account of something if it is to have the assertive content required to serve as justification of other claims
 2. And the observation itself must be justified in the sense of being certifiably not haphazard or uncontrolled.
 - Observations must be *carefully* done and *repeatable*; carried out in a *controlled way and under proper conditions*.
 - This requires an understanding of the relevant conditions under which an observation is done.
 - The scientific community must understand what it takes to do the observation properly if the observation is to be acceptable as scientific evidence.

OBSERVATION

- **Accountable Observation:**

- To be useful as an evidence, an observation must make contact with the world in some way if it is to serve as proof or as refutation.
 - Science can make use of observation only if it is an *accountable* observation:
 1. The observation must be *informative*: it must be an account of something if it is to have the authority required to serve as justification of other claims
 2. And the observation must be justified in the sense of being certifiably not haphazard.
- In short: To contribute to science, an observation must be informative, assertive, propositional.*
- It must be *carefully* done and *repeatable*; carried out in a *controlled way* and under *relevant conditions*.
- This requires an understanding of the relevant conditions under which an observation is done.
- The scientific community must understand what it takes to do the observation properly if the observation is to be acceptable as scientific evidence.



OBSERVATION

- All observation in science is influenced by theory:
 - Informative observation is impossible without a *theoretical background*, and
 - *Alternative theoretical backgrounds* lead to *alternative informational reports*.

*Observation without
theory is blind!*

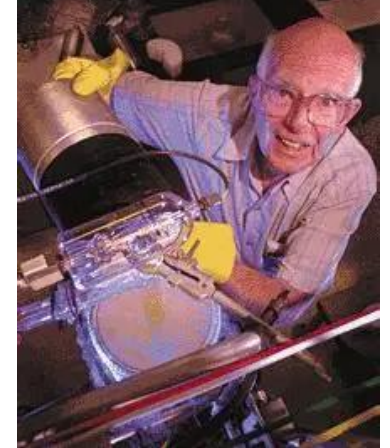
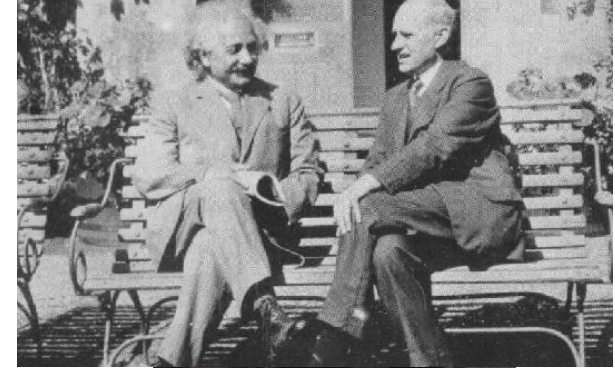
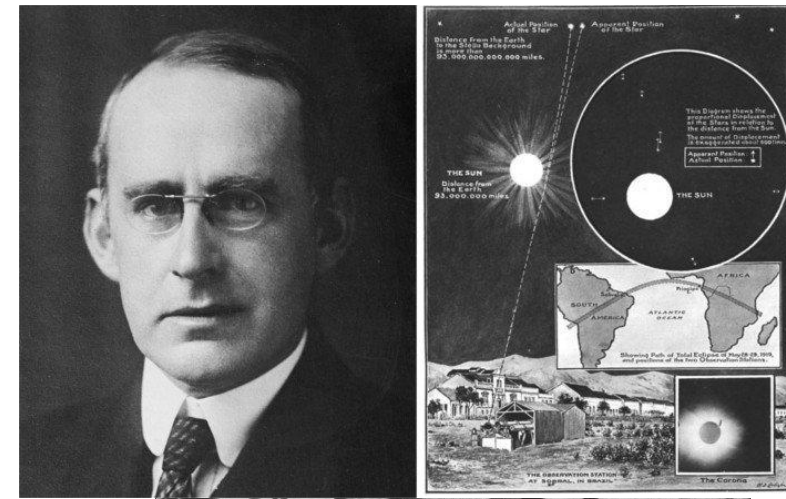


Immanuel Kant
1724-1804



OBSERVATION

- Theory influences observation in several ways:
 - ❖ Theoretical understanding of the world guides our **selection** of **what** observations to do.
 - Example:*
 - Einstein's theory indicated that an observation of stars during an eclipse would be informative.
 - Solar theory directs observational efforts toward the search for neutrinos.
 - This is, of course, the way things have to be and ought to be if the idea is to test a theory by its consequences.
 - ❖ Different theories precipitate a different collection of evidence.





OBSERVATION

- Theory influences observation in several ways:
 - ❖ Theoretical understanding of the world guides our **selection** of **what** observations to do.
 - ❖ Theories guide our decisions about when to **count** observations as **credible**.
 - Theories supply standards to *evaluate the reliability* of observational reports
 - The assessment of the viewing conditions, such as adequate lighting; attesting that there are no distorting factors, etc., is largely based on an understanding of the *causal mechanisms* of observation.
 - The role of theory is especially apparent in cases of observations that are assisted by machines.
 - A great amount of theory goes into the building of the machines (e.g., telescope, microscope, cloud chamber), and into sanctioning their reliability.
 - The data is acceptable, that means, the observations are accountable, only if the experts, who invent and build the machines, have a solid theoretical account of *how they work*.



OBSERVATION

- Theory influences observation in several ways:
 - ❖ Theoretical understanding of the world guides our ***selection*** of ***what*** observations to do.
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Accounting theory:

A theoretical claim used to attest to the reliability of an observation or to describe the informational content of an observational claim is being used as an accounting theory.

It is a designation of how a theory is used rather than of any intrinsic features of content or scope of a theory.

As an example, theoretical claims about the behavior of electrons are used as accounting theories in the case of observing a DNA molecule (or anything else) with an electron microscope.



OBSERVATION

- Observations in science are theory-laden
 - They always depend on the theories we use to *interpret* them.
 - *Example:* To say that the “Sun rises in the east every day”, involves an understanding of the concept of “Sun” and “east” as well as geographical and astronomical knowledge.
 - This dashes the empiricist’s hope of a common ground of evidence agreeable to everyone regardless of his or her theoretical predisposition.
- ***Observations are beliefs we get as a result of direct impact by the environment, though they gain their importance as evidence to function in the process of science only with reference to theory.***

EARLIER!

OBSERVATION



- Reading ***the book of nature*** – a metaphor for doing natural science – requires observing the events in nature.
- To have any claim to an accurate report of what's going on in ***this*** world, there must be ***input*** from the world.
- Observation is, thus, a dietary staple of empirical science.



OBSERVATION

- When we read a book, we want to know the plot and the meaning of the text
 - We want to learn what is beyond the manifest *marks* on the pages – we want to understand what the *marks* indicate and what they mean.
- Science is doing the same!
 - Science aims to understand what the *observations* indicate, and this is done only through a careful attention to the observations themselves.
 - This includes forming concepts of things so that tokens of sensation can be grouped into interacting kinds of things and the relevant kinds can be individuated and understood.
 - Thus, observations are meaningful contributions to our understanding of nature only with some claims of connection between the image in sensation and the specimen in the world.

BLURRING THE INTERNAL- EXTERNAL DISTINCTION

BLURRING THE INTERNAL-EXTERNAL DISTINCTION



Reflect & Discuss

6. “It seemed natural to begin by dividing the fund of information that could be used to evaluate a theory in two categories, depending on the source of the information.”

What two categories are being alluded to here?

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- So far, we've distinguished between the internal and external features of a theory:
 - *Internal features*: i.e., information from theories themselves, both the theory being evaluated and other theories relevant to it.
 - E.g., entrenchment, explanatory cooperation
 - *External features*: i.e., information from the world.
 - E.g., explanation, observation, confirmation

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- However, as we've seen, the distinction between internal and external features is not so sharp. Instead, it is rather *blurry*:
 - E.g., observations themselves bear a theoretical component (see slides earlier).



OBSERVATION

- Theory influences observation in several ways:
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BLURRING THE INTERNAL-EXTERNAL DISTINCTION



Reflect & Discuss

7. “The ambition of science, to describe more than is apparent, keeps it steadily at risk of being wrong ... Our situation with respect to the unobservable world of scientific images has us, as perceivers, at one end of a causal chain ...”

How exactly are we stuck?

- Each theoretical claim is a declarative assertion about the world, and the business of justification is the issue of truth (*how likely it is that the claim is true*).
 - However, there is no direct access to information on the truth and falsity of the theoretical claims made.

- Each theoretical claim is a declarative assertion about the world, and the business of justification is the issue of truth (*how likely it is that the claim is true*).
 - However, there is no direct access to information on the truth and falsity of the theoretical claims made.
 - This is particularly true when it comes to *unobservables*: We don't have direct access to them. All we have is evidence of their effects (e.g., cloud chamber images).
 - Effects are informative of their causes only with the assistance of a causal theory. That is, claims about the unobservable causes are products of inference. And the inference must be sanctioned by a theoretical conditional of the form: *If the observable effect has feature x, then the unobservable cause has feature y.*
 - And even evidential claims about *observables* don't provide us with the truth since they simply report how things are in the world, and are subject to justification and are revisable under pressure from theories.

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- The process of justification in science must acknowledge this predicament.
- It must pay attention to what information is available to us and what is in principle beyond the limits of awareness.
- Only the available information can be relied on to guide the justification and to supply the proof of theories.

BLURRING THE INTERNAL-EXTERNAL DISTINCTION



Reflect & Discuss

8. “Much as we would like to use correspondence with-the-facts as the measure of justification, we can’t. We simply have no direct way of telling when correspondence has been achieved. We are stuck with coherence ...”

How exactly are we stuck?

Also, read the two passages on your worksheet

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- Justification is to be a symptom of truth:
 - A properly designed process of justification will allow us to say that if a belief, theory, is justified then it *is likely to be true*.

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- Justification is to be a symptom of truth:
 - The justification will take place either by comparing the theory to the world to see if the description *corresponds* to the facts or by *comparing* the theory to other theories to see if it fits into the theoretical system and *coheres* with the web of theories.

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- Justification is to be a symptom of truth:
 - The justification will take place either by comparing the theory to the world to see if the description *corresponds* to the facts or by *comparing* the theory to other theories to see if it fits into the theoretical system and *coheres* with the web of theories.
- Correspondence is the nature of truth, but it is exactly this relation of correspondence that we are unable to evaluate.
- We are stuck with coherence, judging theories by their relation to other theories.
 - It is **not** that *coherence is* truth; rather, *coherence indicates* truth (it indicates correspondence).

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- As science is a communal undertaking, the process of justification must be internal to the community of investigators.

BLURRING THE INTERNAL-EXTERNAL DISTINCTION

- As science is a communal undertaking, the process of justification must be internal to the community of investigators.
- Note that this doesn't mean that science is thoroughly subjective:
 - There are still standards, internal standards as we've discussed earlier in the course (*recognizable*)
 - Science uses objective statements to *accurately* describe objects.
 - Science is also done objectively insofar as it is done with an openness to discussion and a willingness to hear and consider objections to any theory (*discussible and accountable*)

In-Class Activity

In-Class Activity

- Please create a poster of the assigned chapter. It should include the important concepts and ideas of the chapter. You will present it in the next class (max. 5 mins per group).
- Please write a list (handwritten or a file) that contains keywords and their definition (send it to me via e-mail by Friday)
- Both will help you prepare for the final exam.

In-Class Activity

Chapter	Group(s)
Chapter 1	Magscienticks
Chapter 2	Aster, Group #1
Chapter 3	Failed Experiments, Hoomans
Chapter 4	The Quarks, Trial & Error
Chapter 5	Trinity, Among Us
Chapter 6 + 7	Meow Meow